

Imran Jahid Khan

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🌐 ijk37.com • [Google Scholar](https://scholar.google.com/citations?user=ijk37) • [id 0009-0007-2666-3354](https://orcid.org/0009-0007-2666-3354) • 📍 Tuskegee, AL, USA

Research Interests

Security and Resilience of Cyber-Physical Systems • Sensor Attacks, Anomaly Detection, and Recovery • Robust Reinforcement Learning for Control • Cloud-Controlled and Medical CPS • Blockchain-Based Integrity Verification

Education

Tuskegee University, Alabama, USA

Aug 2025 – May 2027 (expected)

M.S. in Information Systems and Computer Security

Advisor: Dr. Mohammad Arafatur Rahman

Relevant Coursework: Information Security; Network Security and Management; Data Networks & Cloud Computing; Data Mining & Machine Learning; Statistics with R; Data Analytics; Big Data Analytics; Database Systems.

Bangladesh University of Engineering & Technology (BUET), Dhaka

2011 – 2016

B.Sc. in Mechanical Engineering

Thesis: Wind Turbine Blade Design for Optimum Power Output.

Mathematical Foundations: Calculus; Linear Algebra; Ordinary & Partial Differential Equations; Complex Variables & Fourier Analysis; Numerical Analysis.

Research Experience

Graduate Research Assistant

Aug 2025 – Present

Dept. of Computer Science, Tuskegee University • Advisor: Dr. Mohammad Arafatur Rahman

- Model sensor-layer attacks (replay, false-data injection, Gaussian observation corruption) on learning-based CPS controllers and build detection-and-recovery defenses that keep closed-loop control stable in simulation.
- Built cloud-controlled reinforcement-learning CartPole testbeds (Q-learning, DQN, PPO); showed that corrupting a single critical state variable (pole angle) degrades control as severely as corrupting the full state, so defenses must be targeted rather than uniform [2]. Developed LSTM-based integrity-attack detection for cloud-based CPS [1].
- Paired a PPO controller with an LSTM-autoencoder anomaly detector and a blockchain-anchored recovery mechanism that deterministically detects replay attacks and restores the most recent verified clean state [3].
- Designed reconstruction-assisted recovery for artificial-pancreas RL control under CGM sensor attacks, combining a residual-LSTM state reconstructor with a safety shield; evaluated glucose-control and risk metrics in the simglucose simulator [4].
- Evaluated classical and quantum RL controllers under meal-information uncertainty in artificial-pancreas systems [5]; contributed to an auditable evidence-ranking framework for AI-assisted NMR metabolomics [6].

Publications

PUBLISHED CONFERENCE PAPERS

[†] presenting author

- [1] M. A. Rahman, I. J. Khan, and F. Wu, "Detection of Integrity Attacks in Cloud-Based Cyber-Physical Systems (CBCPS) Using Long Short-Term Memory (LSTM) Models," in *Proc. 2026 9th International Conference on Information and Computer Technologies (ICICT)*, pp. 533–538, 2026.
- [2] I. J. Khan, M. Hossain, M. A. Rahman, and F. Wu, "Resilience Analysis of Cloud-Controlled Cyber-Physical Systems Using a Reinforcement Learning CartPole Testbed," in *Proc. 2026 IEEE World AI IoT Congress (AllIoT)*, pp. 773–779, 2026.

ACCEPTED CONFERENCE / WORKSHOP PAPERS

- [3] I. J. Khan[†], M. Hossain, M. A. Rahman, and F. Wu, "Blockchain-Anchored Detection and Recovery for Resilient Reinforcement Learning-Based Cyber-Physical Systems," *7th Annual National HBCU Blockchain, FinTech, and AI Conference*, National FinTech Center, Morgan State University, 2026.
- [4] I. J. Khan[†], S. Tahmim, M. Hossain, M. A. Rahman, and F. Wu, "Reconstruction-Assisted Resilient Reinforcement Learning for the Artificial Pancreas under CGM Sensor Attacks," *Proc. IEEE International Conference on E-health Networking, Application & Services (HealthCom) Workshops, ISTHS Workshop*, 2026.
- [5] M. Hossain, I. J. Khan[†], M. A. Rahman, and F. Wu, "Evaluating Classical and Quantum Reinforcement Learning Controllers under Meal-Information Uncertainty in Artificial Pancreas Systems," *Proc. IEEE HealthCom Workshops, ISTHS Workshop*, 2026.
- [6] S. Tahmim, I. J. Khan, and W. E. Collier, "An Auditable Evidence-Ranking Framework for AI-Assisted NMR Metabolomics in Breast Cancer Research," *Proc. IEEE HealthCom Workshops, ISTHS Workshop*, 2026.

- [7] **I. J. Khan**, M. Hossain, M. A. Rahman, and F. Wu, "Reconstruction-Assisted DQN for Resilient Cloud-Controlled Cyber-Physical Systems against Data Integrity Attacks," submitted to *2nd Special Session on Cybersecurity and Telecommunication in the Era of AI, 2026 IEEE International Conference on Big Data (BigData)*, 2026.
- [8] M. Hossain, **I. J. Khan**, M. A. Rahman, and F. Wu, "Evaluating Learning Efficiency and Computational Trade-Offs in Classical and Quantum Reinforcement Learning Using a CartPole Testbed," submitted to *2026 IEEE International Conference on Big Data (BigData)*, 2026.

Presentations

- **Oral, Nov 8–10, 2026 (scheduled):** paper [3], 7th Annual National HBCU Blockchain, FinTech, and AI Conference (National FinTech Center, Morgan State University), Nashville, TN.
- **Oral, Oct 2026 (scheduled):** papers [4] and [5], ISTHS Workshop, IEEE HealthCom 2026.
- **Poster, 2025:** "Evaluating Q-Learning Rewards in Noisy Cloud-Based Cyber-Physical Systems." *AWS-MLU AI/ML Fall Teaching and Research Symposium & Student Poster Session*.

Honors & Scholarships

- **Full Scholarship (Graduate Assistantship)**, M.S. in Information Systems and Computer Security, Tuskegee University, 2025
- **Japan Human Resources Development Scholarship (JDS)**, Government of Japan, for graduate study at Hiroshima University, 2025 (*declined*)
- **Full Scholarship**, M.A. in Policy Economics, Williams College, 2025 (*declined*)
- **Bangladesh Civil Service (Administration) Cadre**, 37th BCS Examination: merit position 89 of 306 appointees from about 243,000 candidates
- **Prime Minister's Fellowship**, Government of Bangladesh, 2024
- **Foreign Master's Scholarship**, "Capacity Building of BCS Administration Academy (2nd Revised)" Project, Government of Bangladesh, 2024 (*declined*)

Teaching Experience

Graduate Teaching Assistant

Introduction to Computer Science, Dept. of Computer Science, Tuskegee University

Aug 2025 – Dec 2025

Professional Experience (Bangladesh Civil Service)

Deputy Director (Senior Assistant Secretary)

Export Promotion Bureau, Ministry of Commerce

Sep 2024 – Jul 2025

- Produced data-informed export-policy recommendations and worked with stakeholders to digitize trade-facilitation processes.

Assistant Commissioner (Land) & Executive Magistrate

Upazila Land Office, Patuakhali & Bhola, Ministry of Land

Nov 2022 – Aug 2024

- Managed digital land records and reporting to support land administration; adjudicated land disputes and coordinated field operations.

Assistant Commissioner & Executive Magistrate

Office of the Deputy Commissioner, Cox's Bazar, Ministry of Public Administration

Apr 2019 – Nov 2022

- Coordinated disaster-relief and COVID-19 response using ICT; administered public budgets, personnel, and electronic procurement.

Technical Skills

Programming: Python, C/C++, R, SQL, MATLAB

ML / RL: PyTorch; Q-learning, DQN, PPO; LSTM and LSTM autoencoders; anomaly detection; state reconstruction

Simulation: Gymnasium (CartPole control testbeds), simglucose (artificial-pancreas simulator)

Tools: Git/GitHub, Linux, LaTeX

Standardized Tests

GRE: 322 (Quantitative 167), Dec 2024

IELTS Academic: 7.0, Jan 2024